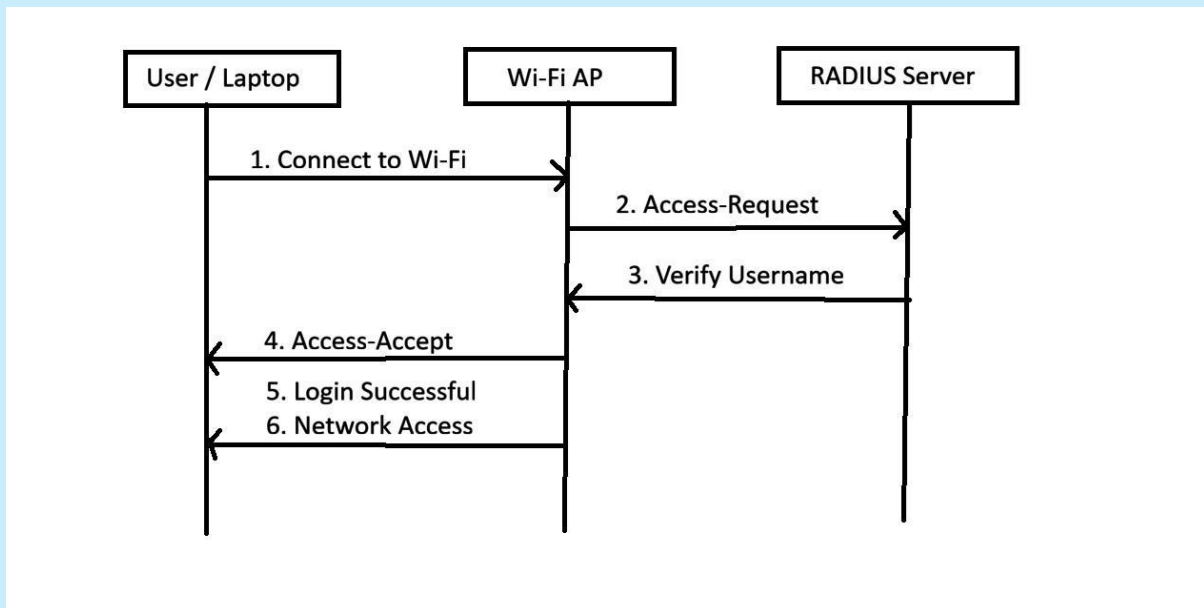


# **RADIUS & AMP**

**1. Create a comparison table listing at least 4 key differences between RADIUS and TACACS+ features, focusing on protocol type, encryption, ports used, and command authorization.**

| Feature                         | RADIUS                                                                                                    | TACACS+                                                                                                          |
|---------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>1. Protocol Type</b>         | Uses <b>UDP</b> protocol.                                                                                 | Uses <b>TCP</b> protocol.                                                                                        |
| <b>2. Encryption</b>            | Encrypts mainly the <b>user password</b> ; other information is generally not encrypted.                  | Encrypts the <b>entire packet payload</b> , providing better protection for authentication data.                 |
| <b>3. Ports Used</b>            | <b>UDP 1812</b> for authentication and <b>UDP 1813</b> for accounting.                                    | <b>TCP 49</b> for authentication, authorization, and accounting.                                                 |
| <b>4. Command Authorization</b> | Provides limited authorization and generally does not provide detailed <b>per-command authorization</b> . | Supports detailed <b>command-by-command authorization</b> , especially useful for network device administration. |
| <b>5. AAA Functions</b>         | Authentication and authorization are combined, while accounting is separate.                              | Authentication, authorization, and accounting are <b>separated</b> .                                             |
| <b>6. Common Use</b>            | Commonly used for <b>Wi-Fi, VPN, and network access authentication</b> .                                  | Commonly used for <b>Cisco/network device administrator access</b> .                                             |

**2. Draw a simple flowchart or sequence diagram that illustrates the RADIUS authentication process for a user trying to log in to a Wi-Fi network (like at a coffee shop or college campus).**



1. User connects to Wi-Fi – The student enters username and password.
2. Wi-Fi AP sends Access-Request – The Access Point forwards the authentication request to the RADIUS server.
3. RADIUS verifies credentials – The RADIUS server checks the username/password.
4. Access-Accept – If credentials are correct, RADIUS sends approval to the Wi-Fi AP.
5. Login successful – The AP allows the user to connect.
6. Network access granted – The user can access the college/coffee-shop network.

**3. Describe a real-world scenario from any app you use (such as Zomato, Paytm, or IRCTC) where TACACS+ would be more suitable than RADIUS for managing admin access, and explain why.**

### **Real-World Scenario – TACACS+ for Zomato**

Consider a company like Zomato that has many servers, routers, switches, and firewalls. Network administrators need to log in to these devices to configure and manage the network.

In this situation, TACACS+ would be more suitable than RADIUS for managing administrator access.

### **Why?**

- TACACS+ separates authentication, authorization, and accounting (AAA).
- It can provide command-level authorization, so an administrator can be allowed to run some commands while being restricted from running sensitive commands.

- It encrypts the entire TACACS+ packet payload, providing better protection for administrative communication.
- It provides detailed accounting/logging, making it easier to track which administrator performed which action.

### Example:

A Zomato network administrator may be allowed to view network configurations but not change firewall rules. TACACS+ can control this at the command level.

## 4. List two advantages and two disadvantages of using RADIUS for network authentication in a public hotspot (like those found in malls or airports).

### Advantages:

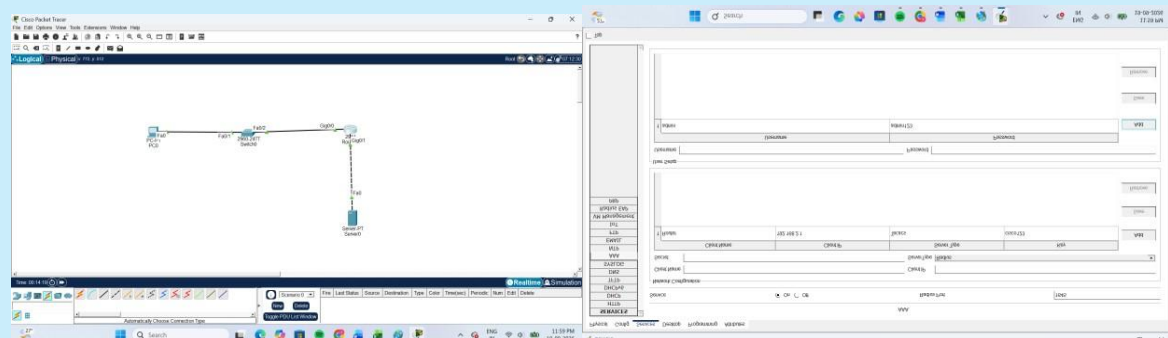
1. Centralized Authentication: RADIUS allows the hotspot to manage user authentication from a central server instead of storing user credentials on every access point.
2. Scalable: It can handle a large number of users, making it suitable for public Wi-Fi in malls, airports, hotels, and colleges.

### Disadvantages:

1. Server Dependency: If the RADIUS server becomes unavailable, users may not be able to authenticate and access the Wi-Fi network.
2. Security Configuration Required: RADIUS security depends on proper configuration, such as using strong shared secrets and secure authentication methods. Poor configuration can expose user credentials or authentication traffic.

## 5. Use ChatGPT or another AI tool to generate a sample TACACS+ configuration for a Cisco router, then review the output and highlight any commands you would modify for a college lab setup.

**Hint:** Ask the AI tool for 'basic TACACS+ configuration for Cisco router' and check for lines like 'tacacsserver host' and 'aaa new-model'.



```
PCO
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 192.168.1.1: bytes=32 time=0ms TTL=255
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ssh -l admin 192.168.1.1

Password:
#>
```